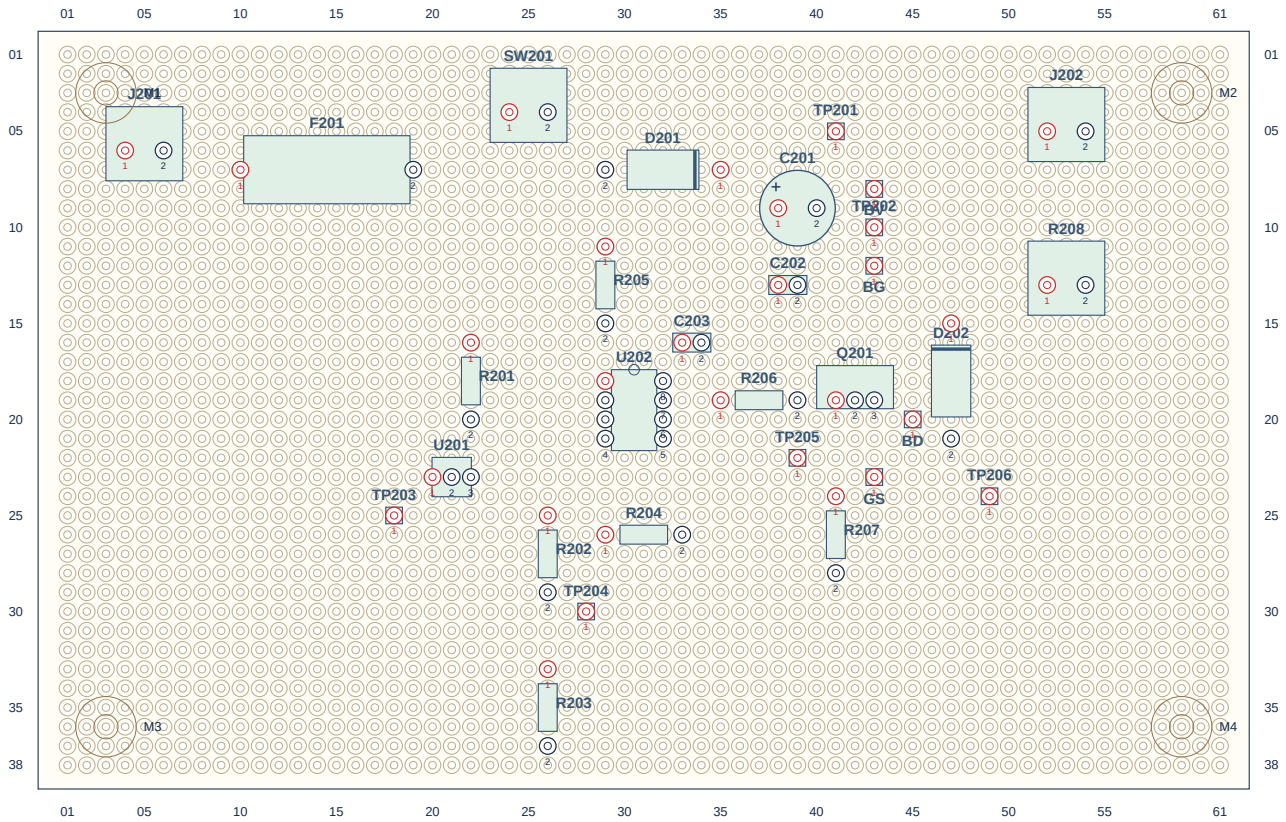
# ONE-SERVO protected supply: component placement

Component side; copper underneath. SW201 and R208 are cable interfaces; actual parts mount off-board.

P1 - UNBUILT / FIRST BUILD = ONE SERVO




**ASSEMBLY REFERENCES**

J201: 1 PSU+ / 2 GND

J202: 1 SERVO_VS / 2 GND

SW201: two-wire switch harness

R208: external 2.2-ohm / HS25

C201: 1000uF/25V D10 x H20

C201: 5.00-mm leads -> 5.08 mm

F201: ESKA 503.400 + 503.310

F201: T2.5A fuse, provisional

U201: ST TL431CZ REF1 / A2 / K3

U202: LM393 DIP8, notch at top

Q201: IRLZ44NPBF G1 / D2 / S3

Q201: metal tab = DRAIN

D201/D202: 1 cathode / 2 anode

BV/BG/BD/GS: existing-pad nodes

**DRILL / FIT NOTES**

Terminals + Q201: 1.3-mm holes

F201 + D201/D202: 1.5-mm holes

F201: 22.60 -> 22.86-mm pitch

Dry fit; never force holder leads.

No copper tracks: hand-wire with insulated conductors. Off-board load resistor needs a heatsink.

Place hot resistor and MOSFET clear of capacitor and wiring. Physical/thermal build is unverified.

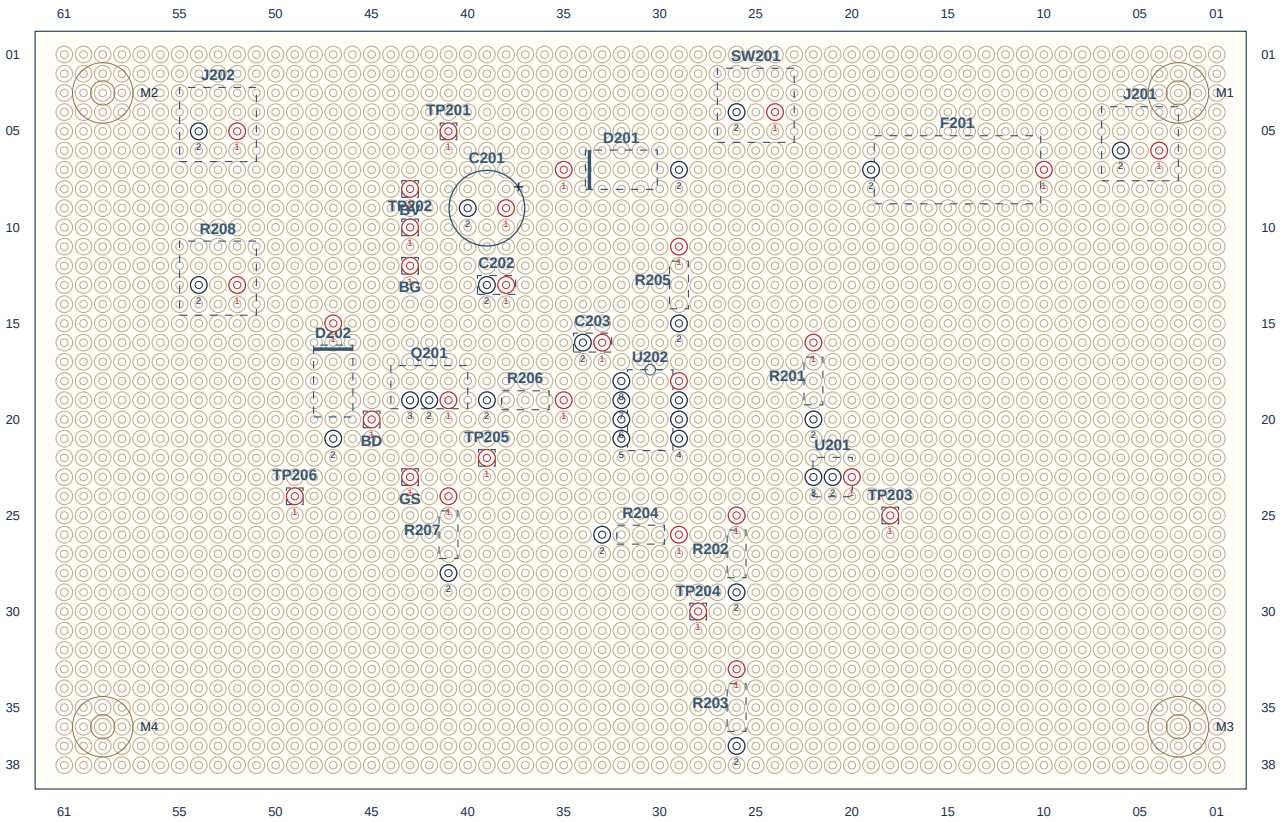
**Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.**

50 mm scale check

# ONE-SERVO protected supply: solder-side reference

Mirrored horizontally: column61 left, column01 right. Rows and physical hole names are unchanged.






**SOLDER-SIDE ORIENTATION**

Flip like turning a page.

Do not mirror again.

TP201: SERVO_VS

TP202: GND

TP203: VREF

TP204: SENSE

TP205: MOSFET gate

TP206: MOSFET drain

C201 positive = pad1.

Diode stripe = pad1 cathode.

U202 pin7 stays unconnected.

Keep load wires short and reinforced.

Quiet ground joins at Q201 source.

No load current through IC leads.

Mount reserves are optional.

RE200-LF has no plated-through holes.

Crossing insulated wires do not connect. Only the endpoints in the wire list are soldered together.

Front-side copper rings in KiCad are virtual: actual board has solder pads on its underside only.

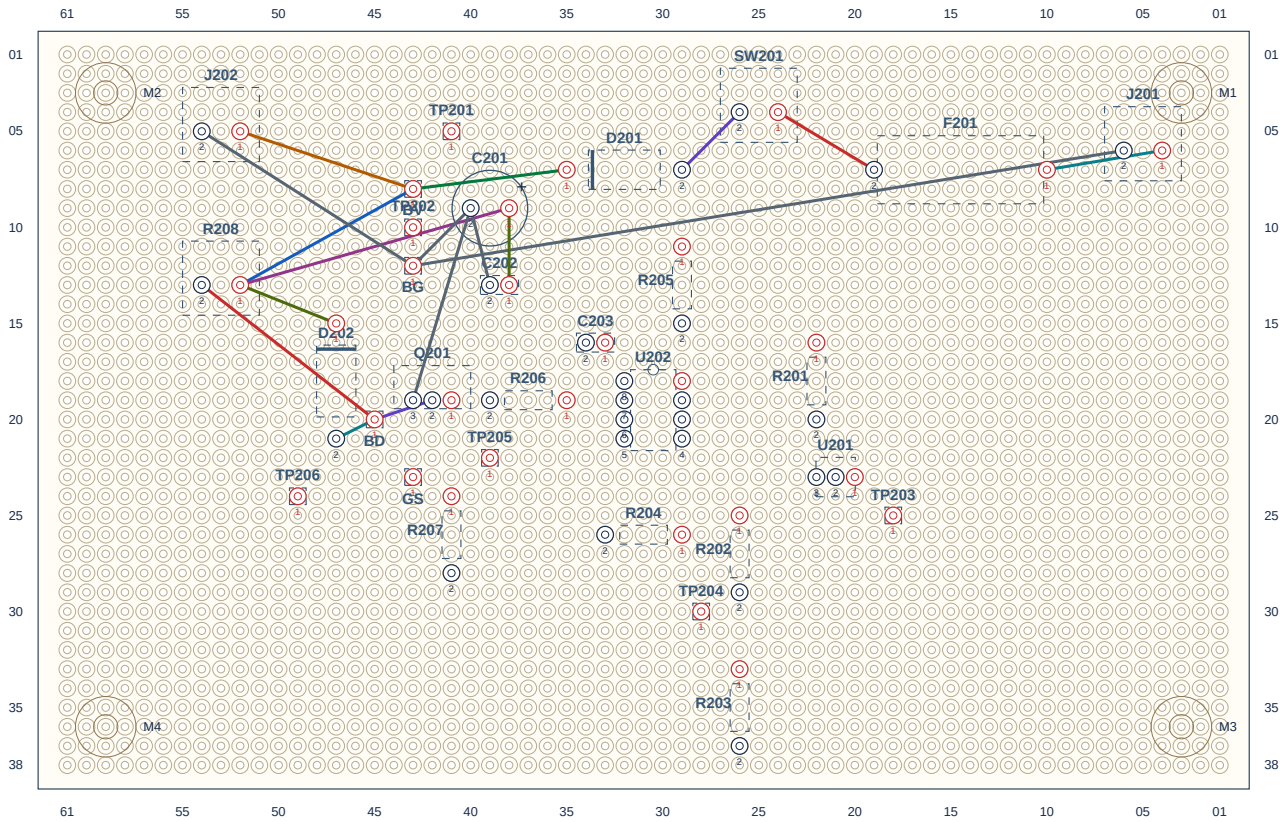
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50 mm scale check

# Solder-side wiring: load-current wiring

Endpoint guide only. Use insulated passovers; check polarity and continuity before energizing.

P1 - UNBUILT / FIRST BUILD = ONE SERVO




| WIRE | FROM / HOLE | TO / HOLE |
|---|---|---|
| W001 | J201.1 C04-R06 | F201.1 C10-R07 |
| W002 | F201.2 C19-R07 | SW201.1 C24-R04 |
| W003 | SW201.2 C26-R04 | D201.2 C29-R07 |
| W004 | D201.1 C35-R07 | BV.1 C43-R08 |
| W005 | BV.1 C43-R08 | J202.1 C52-R05 |
| W006 | BV.1 C43-R08 | R208.1 C52-R13 |
| W007 | R208.1 C52-R13 | C201.1 C38-R09 |
| W008 | D202.1 C47-R15 | R208.1 C52-R13 |
| W009 | D202.2 C47-R21 | BD.1 C45-R20 |
| W010 | BD.1 C45-R20 | R208.2 C54-R13 |
| W011 | BD.1 C45-R20 | Q201.2 C42-R19 |
| W012 | J201.2 C06-R06 | BG.1 C43-R12 |
| W013 | BG.1 C43-R12 | J202.2 C54-R05 |
| W014 | BG.1 C43-R12 | C201.2 C40-R09 |
| W015 | C201.2 C40-R09 | Q201.3 C43-R19 |
| W016 | C202.1 C38-R13 | C201.1 C38-R09 |
| W017 | C202.2 C39-R13 | C201.2 C40-R09 |

Load wires: rated insulated conductor, reinforced at pad nodes. Short resistor/clamp loop.

First build only: one servo, 5.0-5.3V rail target. Clamp behavior and heating require bench checks.

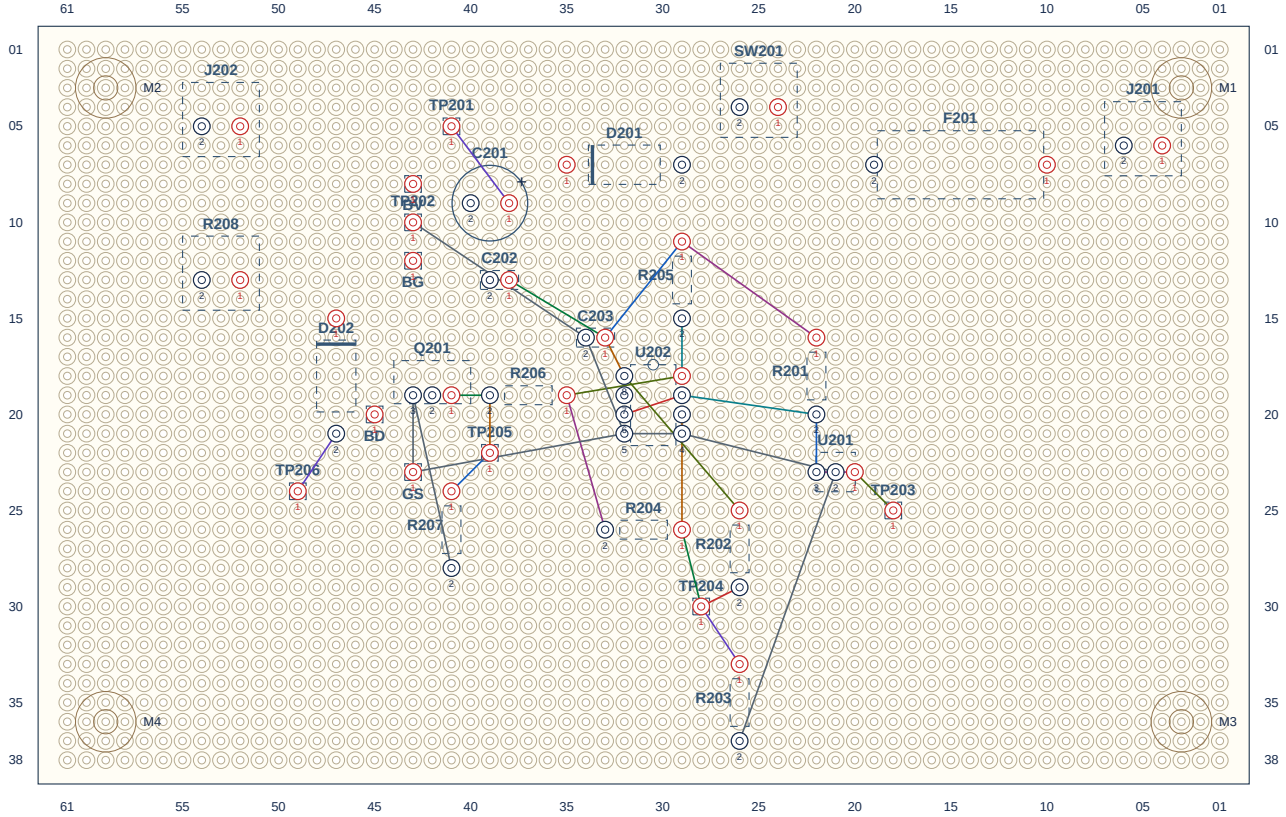
**Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.**

50 mm scale check

# Solder-side wiring: comparator / reference wiring

Endpoint guide only. Use insulated passovers; check polarity and continuity before energizing.

P1 - UNBUILT / FIRST BUILD = ONE SERVO




| WIRE | FROM / HOLE | TO / HOLE |
|---|---|---|
| W018 | Q201.3 C43-R19 | GS.1 C43-R23 |
| W019 | R207.2 C41-R28 | Q201.3 C43-R19 |
| W020 | D202.2 C47-R21 | TP206.1 C49-R24 |
| W021 | Q201.1 C41-R19 | R206.2 C39-R19 |
| W022 | R206.2 C39-R19 | TP205.1 C39-R22 |
| W023 | TP205.1 C39-R22 | R207.1 C41-R24 |
| W024 | R204.2 C33-R26 | R206.1 C35-R19 |
| W025 | R206.1 C35-R19 | U202.1 C29-R18 |
| W026 | U202.1 C29-R18 | R205.2 C29-R15 |
| W027 | R202.2 C26-R29 | TP204.1 C28-R30 |
| W028 | TP204.1 C28-R30 | R203.1 C26-R33 |
| W029 | TP204.1 C28-R30 | R204.1 C29-R26 |
| W030 | R204.1 C29-R26 | U202.3 C29-R20 |
| W031 | R201.2 C22-R20 | U201.3 C22-R23 |
| W032 | U201.3 C22-R23 | U201.1 C20-R23 |
| W033 | U201.1 C20-R23 | TP203.1 C18-R25 |
| W034 | R201.2 C22-R20 | U202.2 C29-R19 |
| W035 | U202.2 C29-R19 | U202.6 C32-R20 |
| W036 | C201.1 C38-R09 | TP201.1 C41-R05 |
| W037 | C202.1 C38-R13 | C203.1 C33-R16 |
| W038 | C203.1 C33-R16 | U202.8 C32-R18 |
| W039 | C203.1 C33-R16 | R205.1 C29-R11 |
| W040 | R205.1 C29-R11 | R201.1 C22-R16 |
| W041 | U202.8 C32-R18 | R202.1 C26-R25 |
| W042 | GS.1 C43-R23 | U202.5 C32-R21 |
| W043 | U202.5 C32-R21 | U202.4 C29-R21 |
| W044 | U202.5 C32-R21 | C203.2 C34-R16 |
| W045 | U202.4 C29-R21 | U201.2 C21-R23 |
| W046 | C203.2 C34-R16 | TP202.1 C43-R10 |
| W047 | U201.2 C21-R23 | R203.2 C26-R37 |

Load wires: rated insulated conductor, reinforced at pad nodes. Short resistor/clamp loop.

First build only: one servo, 5.0-5.3V rail target. Clamp behavior and heating require bench checks.

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50 mm scale check